A4.2 Conservation of biodiversity: data-based questions

NAME	DATE	TOTAL
		_____ / 13

Read the data carefully before you answer. The number of marks shows how many separate points to make. Show your working in calculations.

1 Insect decline and bee food

7 MARKS



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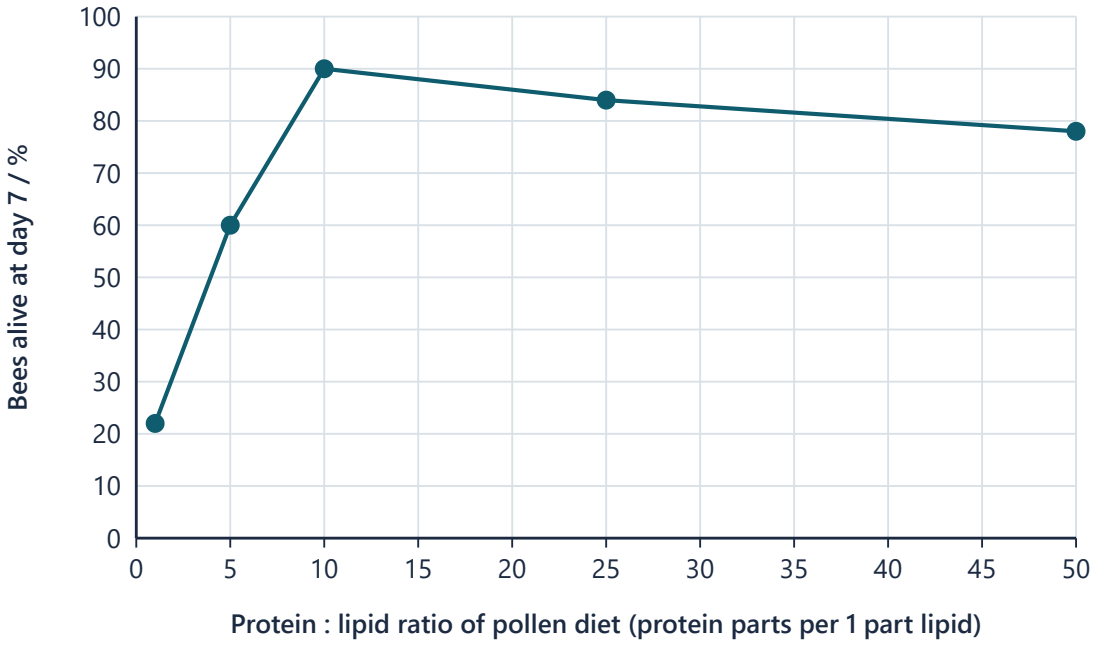
A review of 70 long-term studies estimated how fast insect biomass is falling each year in different orders of insects.

INSECT ORDER	MEAN DECLINE IN BIOMASS / % PER YEAR
LEPIDOPTERA (BUTTERFLIES AND MOTHS)	3.1
COLEOPTERA (BEETLES)	2.4
HYMENOPTERA (BEES, WASPS AND ANTS)	1.9
ALL INSECTS	2.5

a STATE the mean annual decline in biomass for all insects. [1]

b A region has 400 000 insect species. CALCULATE how many would remain after one year if the number of species fell at the same rate. [1]

Bumblebees were fed artificial pollen with different ratios of protein to lipid, and survival was recorded after 7 days.



c DESCRIBE the effect of a high lipid content on survival. [1]

d IDENTIFY, with a reason, the best ratio of protein to lipid for bumblebees. [1]

e DISCUSS how the loss of wildflower habitats could contribute to the decline of bumblebees, using the data. [3]

2 Threats to wilderness

6 MARKS

Large wilderness areas have very few people and are mostly undisturbed. Conservationists recorded which human activities threaten wilderness areas in five biomes. A tick means the activity threatens that biome.

ACTIVITY	TROPICAL FOREST	BOREAL FOREST	TUNDRA	DESERT	GRASSLAND
CLIMATE CHANGE	✓	✓	✓	✓	✓
HUNTING	✓	✓	✓	✓	✓
MINING	✓	✓	✓	✓	
FIRE	✓	✓			✓
LOGGING	✓	✓			
AGRICULTURE	✓				✓
POLLUTION		✓	✓		

a STATE how many of the activities threaten all five biomes. [1]

b **IDENTIFY** one activity that threatens deserts but not grasslands. [1]

c **COMPARE** the threats from pollution and climate change. [2]

d **SUGGEST** how fire could reduce biodiversity in a tropical forest. [2]